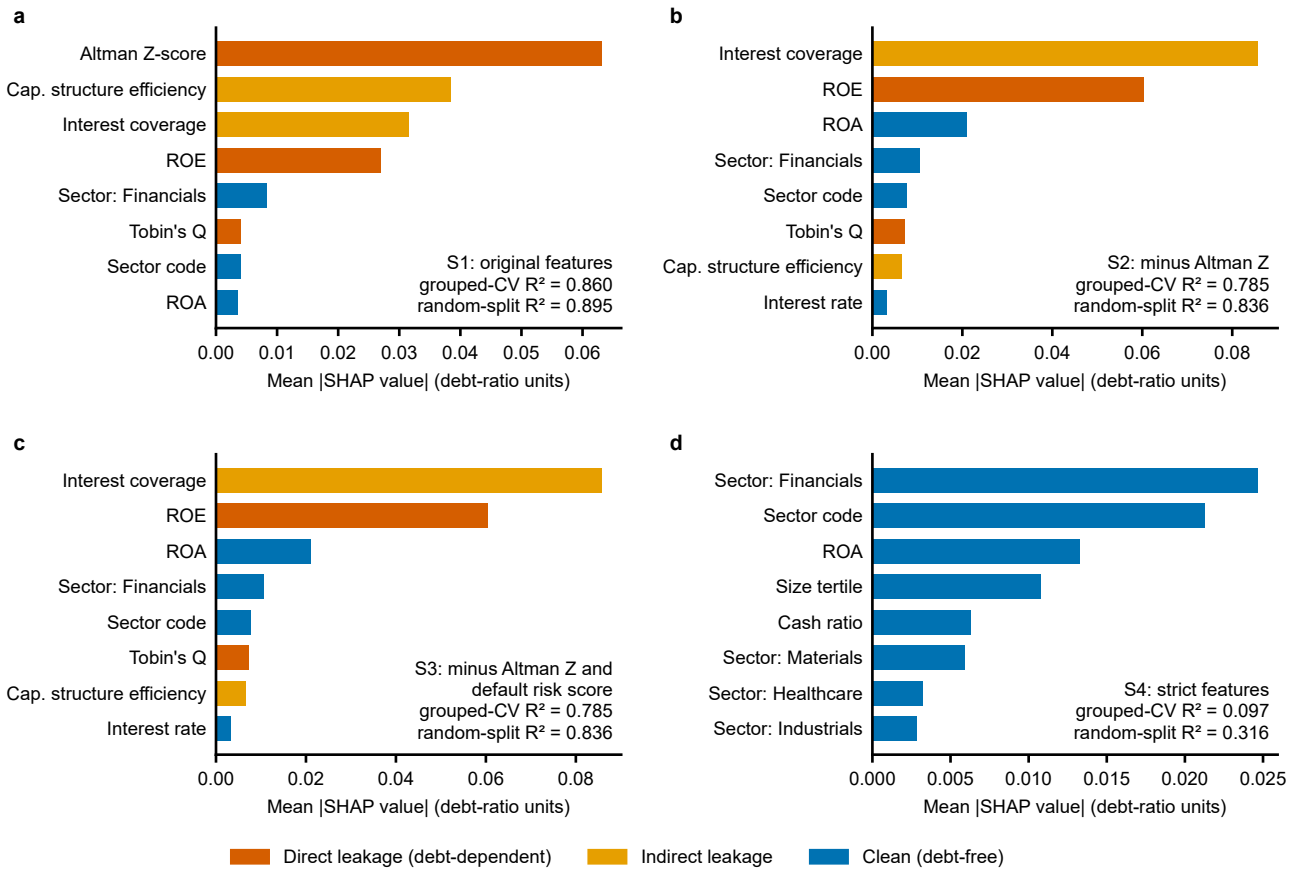
a
Altman Z-score
Cap. structure efficiency
Interest coverage
ROE
Sector: Financials
Tobin's Q
Sector code
ROA
S1: original features
grouped-CV R² = 0.860
random-split R² = 0.895
0.00 0.01 0.02 0.03 0.04 0.05 0.06
Mean |SHAP value| (debt-ratio units)
b
Interest coverage
ROE
ROA
Sector: Financials
Sector code
Tobin's Q
Cap. structure efficiency
Interest rate
S2: minus Altman Z
grouped-CV R² = 0.785
random-split R² = 0.836
0.00 0.02 0.04 0.06 0.08
Mean |SHAP value| (debt-ratio units)
c
Interest coverage
ROE
ROA
Sector: Financials
Sector code
Tobin's Q
Cap. structure efficiency
Interest rate
S3: minus Altman Z and
default risk score
grouped-CV R² = 0.785
random-split R² = 0.836
0.00 0.02 0.04 0.06 0.08
Mean |SHAP value| (debt-ratio units)
d
Sector: Financials
Sector code
ROA
Size tertile
Cash ratio
Sector: Materials
Sector: Healthcare
Sector: Industrials
S4: strict features
grouped-CV R² = 0.097
random-split R² = 0.316
0.000 0.005 0.010 0.015 0.020 0.025
Mean |SHAP value| (debt-ratio units)
Direct leakage (debt-dependent)
Indirect leakage
Clean (debt-free)